

MICHAEL ZHOU

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EDUCATION

Duke University | Pratt School of Engineering

Expected May 2028

B.S. in Electrical and Computer Engineering, B.S. in Computer Science, Minor in Statistics

GPA: 3.9/4.0 | Dean's List (all semesters)

Relevant Coursework: Data Structures & Algorithms, Design and Analysis of Algorithms, Computer Architecture, Linear Algebra, Databases, Distributed Systems, Mathematical Statistics

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C#, Rust, SQL, HTML

Infra & Data: Apache Kafka, Snowflake, AWS (S3, GovCloud), Terraform, Firestore, Datadog

Frameworks: PyTorch, TensorFlow, Flask, React/Native, LangGraph/LangChain, MEAN

WORK EXPERIENCE

DraftKings | C#, Rust, Apache Kafka

June 2026 – Aug 2026

Software Engineering Intern

- Contributed to market making GTM launch in external prediction markets by refactoring pricing-model pipelines and decomposed redundant Kafka topics, improving total throughput by 25% and reducing ~20k lines of legacy tech debt.
- Reduced sports-model's counters API endpoint latency 70% by replacing per-request full-S3 scans with keyed calculationId queries in Snowflake, eliminating production search failures on malformed inputs and improving backtesting reliability.
- Engineered an agentic software catalog for sports-pricing platform, used for on-call engineers, help request and onboarding.
- Refactored mapper logic to emit tombstone events for dropped game states, eliminating 78.7M warning logs per week.

Duke University School of Medicine | PyTorch, MATLAB, CUDA

Nov 2025 – June 2026

Machine Learning Lab Research Assistant

- Automated detection of 20+ rat ultrasonic-vocalization syllables using the DeepSqueak framework and a Faster R-CNN architecture, cutting manual annotation time ~85% while maintaining 98% recall.
- Extracted bioacoustic features (MFCCs, spectral entropy) to distinguish healthy vs. symptomatic groups with 96% accuracy.
- Developed a Python-based preprocessing pipeline to convert raw audio into high-resolution sonograms, implementing adaptive thresholding and Gaussian noise reduction to improve signal-to-noise ratio.

United States Army | C++, Python

May 2025 – Aug 2025

Software Engineering Intern

- Refactored Python and C++ codebase to implement dynamic, anti-jamming channel-hopping capabilities in Crazyflie nano-drones, achieving 95.3% packet delivery ratio under static and dynamic jamming conditions.
- Designed a Bluetooth-based TSCH clock-sync protocol to allow synchronized swarm communication.
- Reduced blackout time by 30% through safe-reinforcement learning risk constraints to bound channel-selection.
- Co-authored JSSR publication on mitigating communication risk by applying safe reinforcement learning constraints.

George Mason University Wireless Cyber Center | C++, Python

Jun 2024 – Aug 2024

Cyber Security Lab Intern

- Engineered a network penetration tool in Python, scripting the Aircrack-ng suite to systematically exploit 802.11 wireless protocols and force controller deauthentication, resulting in an 80% remote takeover rate of target UAVs.
- Developed a remote exploit by reverse-engineering the Bebop Parrot drone's firmware, leveraging an unsecured Telnet service to establish a persistent root-level shell and gain full command-and-control over the drone's operating system.

PROJECTS

Playground AI | [GitHub](#) | LTI 1.3 Application (Python, Vis.js, Flask, Terraform)

2026

- Architected a full-stack Canvas Application procured by Duke OIT and in CI/CD deployment for 1,000+ users.
- Automated application deployments via a CI/CD pipeline and provisioned scalable infrastructure using Terraform (IaC).
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using vector embeddings to parse unstructured PDFs, delivering minimal hallucination, citation-backed answers with Flask and Firestore.
- Developed an unsupervised learning module using K-means clustering to identify curriculum gaps from student queries and built a Knowledge Graph (NetworkX/Vis.js) to visualize semantic relationships between course topics.

Amdahl | [Link](#) | LangChain, ChromaDB, AWS Govcloud

2026

- Built automated document-extraction workflows parsing 10,000+ pages of multi-format government records, cutting manual retention-analysis time by 85% and standardizing metadata pipelines across government agencies.

ADDITIONAL SKILLS & INTERESTS

Fluent in Mandarin; Interested in pickleball and spikeball; Scratch golfer; Love espresso brewing, film photography & traveling